

All India Villages SaaS Platform 🚀

A production-style Full Stack SaaS platform that provides a comprehensive REST API for India's village-level geographical data.

Built using:

- React.js
 - Node.js
 - Express.js
 - PostgreSQL
 - JWT Authentication
 - Recharts
 - Python ETL Pipeline
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Project Overview

This platform provides a centralized API infrastructure for accessing standardized Indian geographical hierarchy data:

Country → State → District → Subdistrict → Village

The system is designed for:

- B2B applications
 - E-commerce platforms
 - Logistics companies
 - Address autocomplete systems
 - Government data services
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Key Features

Authentication & Security

- User Registration
- User Login
- JWT Authentication

- Protected APIs
 - API Key Generation
 - Password Hashing using bcrypt
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✓ **Data Engineering**

- ETL Pipeline using Python
 - Excel Dataset Processing
 - Automated Data Import
 - PostgreSQL Normalized Database
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✓ **Search Engine**

- Real-time Village Search
 - Live Autocomplete
 - Hierarchical Address Results
 - Pagination Support
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✓ **Analytics Dashboard**

- Total States
 - Total Districts
 - Total Subdistricts
 - Total Villages
 - Interactive Charts using Recharts
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✓ **SaaS Architecture**

- REST APIs
- Modular Backend Structure
- Protected Routes
- Scalable Database Design

- API Infrastructure

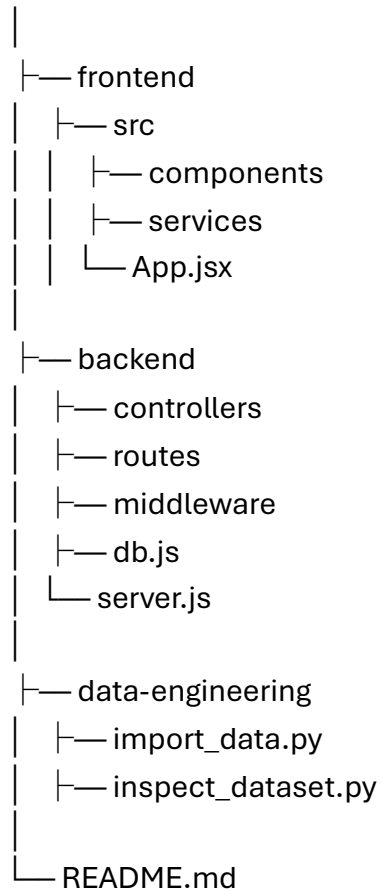
 Tech Stack

Technology Usage

React.js	Frontend
Vite	Frontend Build Tool
Tailwind CSS	UI Styling
Node.js	Backend Runtime
Express.js	API Server
PostgreSQL	Database
Python	ETL/Data Processing
JWT	Authentication
bcryptjs	Password Security
UUID	API Key Generation
Recharts	Data Visualization

Project Structure

Capstone Project



Database Architecture

The platform uses a normalized PostgreSQL schema:

States



Districts



Subdistricts



Villages

Additional tables:

- users
 - api_keys
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Authentication Flow

User Login



JWT Token Generated



Token Stored in LocalStorage



Protected API Access

API Key Infrastructure

Authenticated users can generate API keys for consuming APIs securely.

Example:

POST /api/v1/generate-api-key

Analytics Example

The platform currently supports:

- 28 States
 - 510 Districts
 - 5205 Subdistricts
 - 537,611 Villages
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Search API Example

GET /api/v1/search?q=ram&page=1

Example Response:

```
{
  "results": [
    {
      "village_name": "Rampur",
      "subdistrict_name": "Akkalkuwa",
      "district_name": "Nandurbar",
      "state_name": "MAHARASHTRA"
    }
  ]
}
```

```
]
}
```

ETL Pipeline

The ETL process:

- Reads Excel datasets
- Cleans raw government data
- Normalizes hierarchy
- Imports into PostgreSQL

Implemented using Python and Pandas.

Performance Optimizations

- Indexed Search Queries
 - Pagination APIs
 - Optimized SQL Queries
 - Modular Backend Architecture
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Frontend Features

- Login/Register UI
 - Protected Dashboard
 - Real-time Search
 - Analytics Charts
 - Responsive Design
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Setup Instructions

Clone Repository

git clone <https://github.com/YuvrajHere2002/all-india-villages-saas-platform.git>

Backend Setup

```
cd backend
npm install
node server.js
```

Frontend Setup

```
cd frontend
npm install
npm run dev
```

PostgreSQL Setup

Create database:

```
CREATE DATABASE villages_db;
```

Import schema and datasets using ETL scripts.

Screenshots

Dashboard

- Analytics Cards
- Charts
- Real-time Search

Authentication

- Login Page
- Register Page

APIs

- JWT Protected Routes
 - API Key Generation
-

API Endpoints

Method	Endpoint	Description
POST	/register	Register User
POST	/login	Login User
GET	/analytics	Analytics Data
GET	/search	Village Search
POST	/generate-api-key	Generate API Key

Future Improvements

- Redis Caching
 - Rate Limiting
 - Admin Dashboard
 - NeonDB Deployment
 - Vercel Deployment
 - Advanced Analytics
 - API Usage Tracking
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Learning Outcomes

This project demonstrates:

- Full Stack Development
 - Database Engineering
 - Authentication Systems
 - SaaS Architecture
 - REST API Design
 - ETL/Data Engineering
 - Real-world Problem Solving
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Final Note

This project was developed as a production-style capstone project simulating real-world SaaS platform engineering using large-scale Indian geographical datasets.